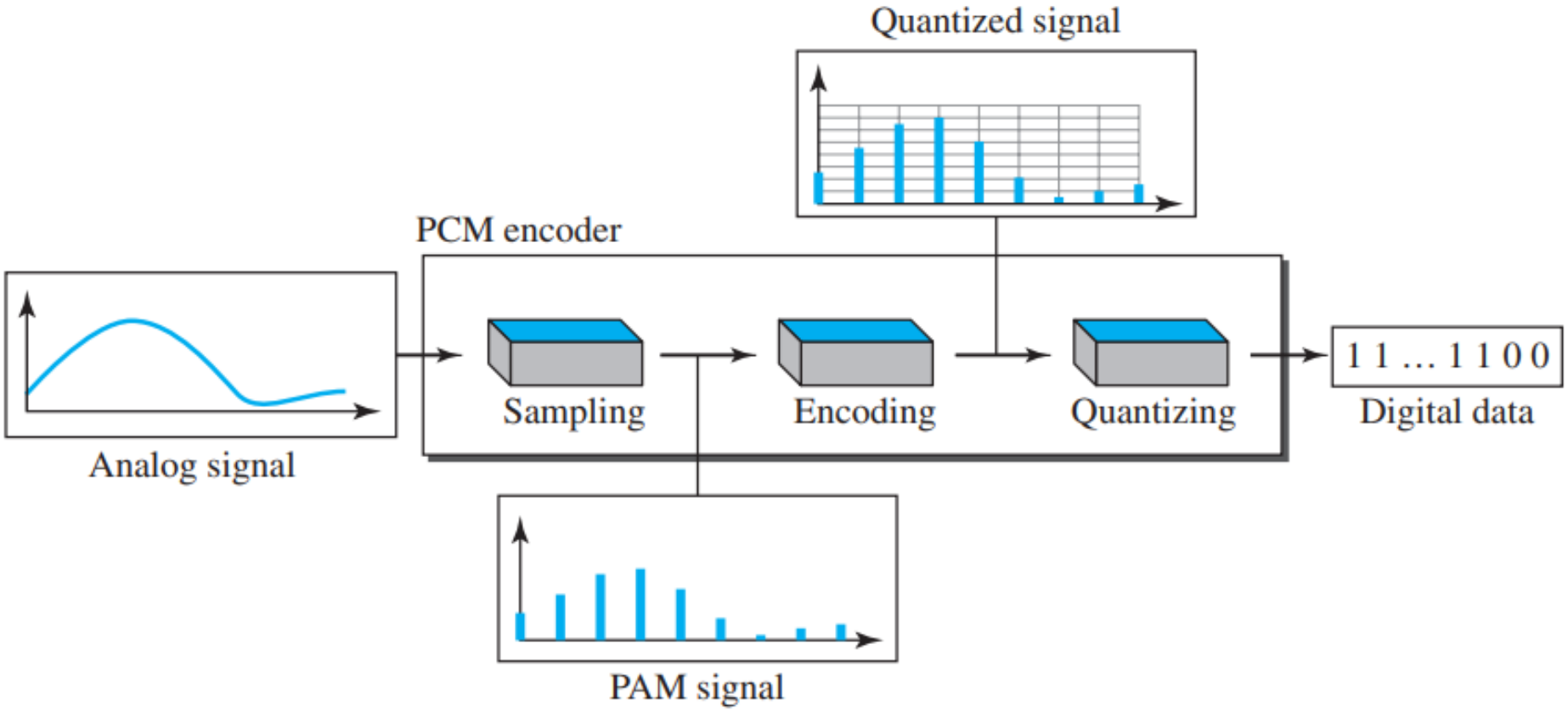
ANALOG-TO-DIGITAL CONVERSION

- Pulse Code Modulation
- Delta Modulation

PULSE CODE MODULATION (PCM)

- The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM).
- A PCM encoder has three processes
  - Sampling
  - Encoding
  - Quantizing

Components of PCM encoder



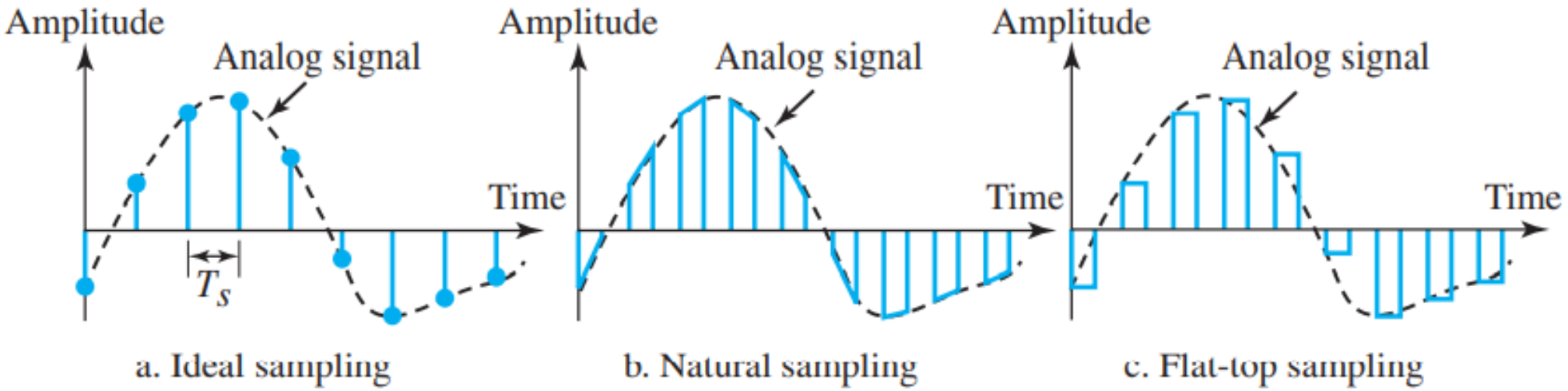
CONTD…

1. The analog signal is sampled.
2. The sampled signal is quantized.
3. The quantized values are encoded as streams of bits.

SAMPLING

- The first step in PCM is sampling.
- The analog signal is sampled every Ts s, where Ts is the sample interval or period.
- The inverse of the sampling interval is called the sampling rate or sampling frequency and denoted by fs, where fs = 1/Ts.
- There are three sampling methods: ideal, natural, and flat-top

THREE DIFFERENT SAMPLING METHODS FOR PCM



a. Ideal sampling

b. Natural sampling

c. Flat-top sampling

CONTD…

- In *ideal sampling*, pulses from the analog signal are sampled.
- This is an ideal sampling method and cannot be easily implemented.
- In *natural sampling*, a high-speed switch is turned on for only the small period of time when the sampling occurs.
- The result is a sequence of samples that retains the shape of the analog signal.

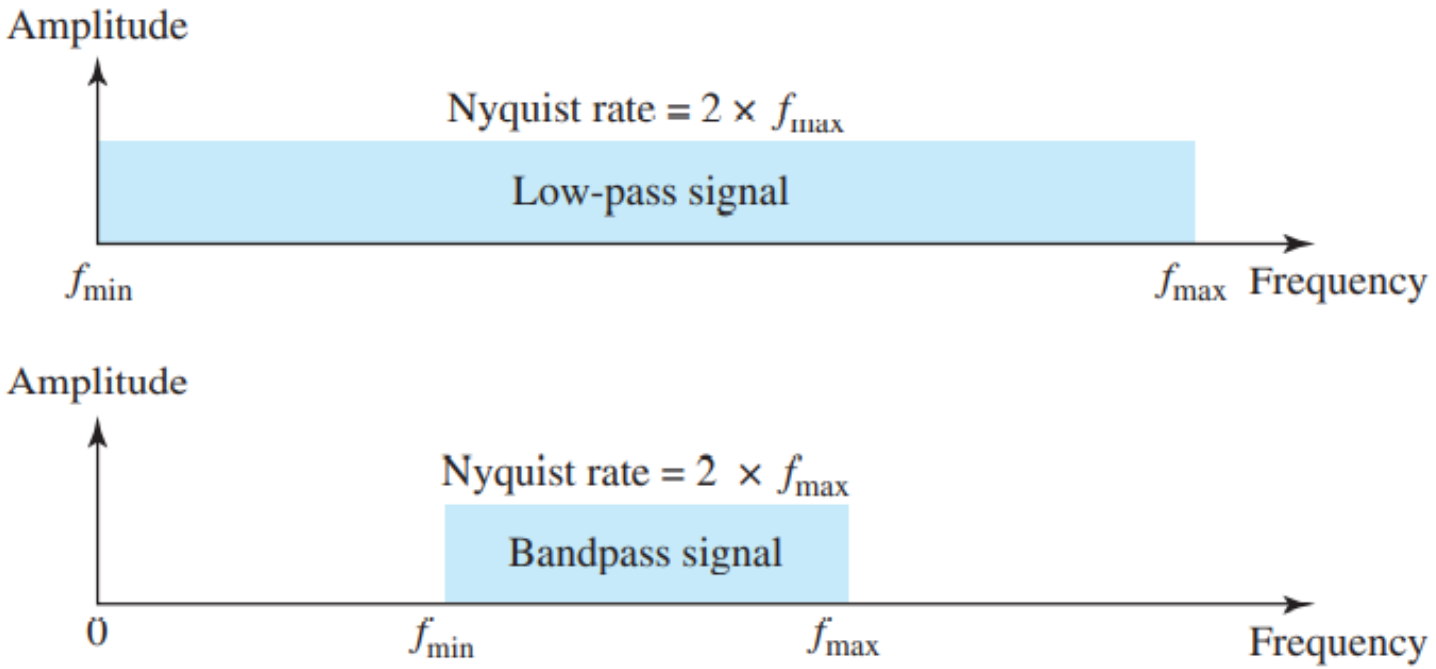
CONTD…

- The Most common sampling method, called sample and hold, however, creates flat-top samples by using a circuit.
- The sampling process is sometimes referred to as pulse amplitude modulation (PAM).

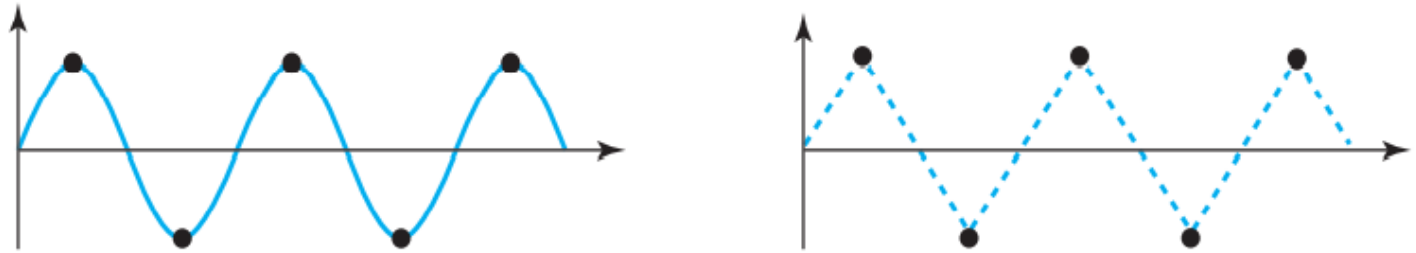
SAMPLING RATE

- One important consideration is the sampling rate or frequency
- According to the Nyquist theorem, the sampling rate must be at least 2 times the highest frequency contained in the signal.
- First, we can sample a signal only if the signal is band-limited.
- Second, the sampling rate must be at least 2 times the highest frequency, not the bandwidth.
- If the analog signal is low-pass, the bandwidth and the highest frequency are the same value.
- If the analog signal is bandpass, the bandwidth value is lower than the value of the maximum frequency.

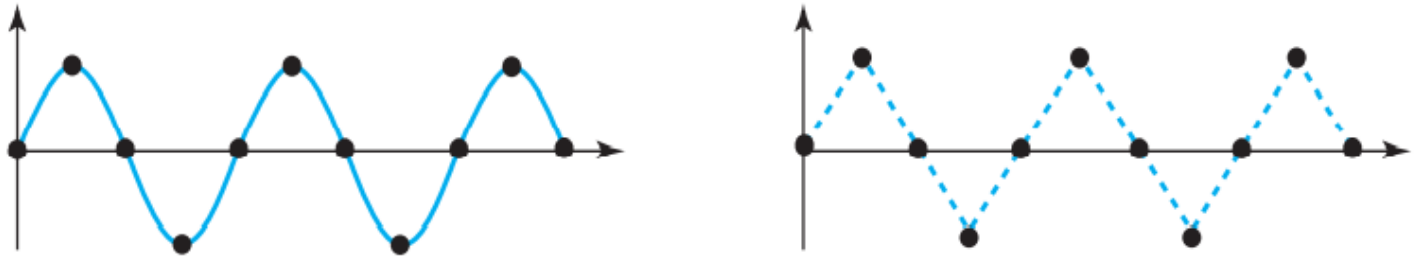
NYQUIST SAMPLING RATE FOR LOW-PASS AND BANDPASS SIGNALS



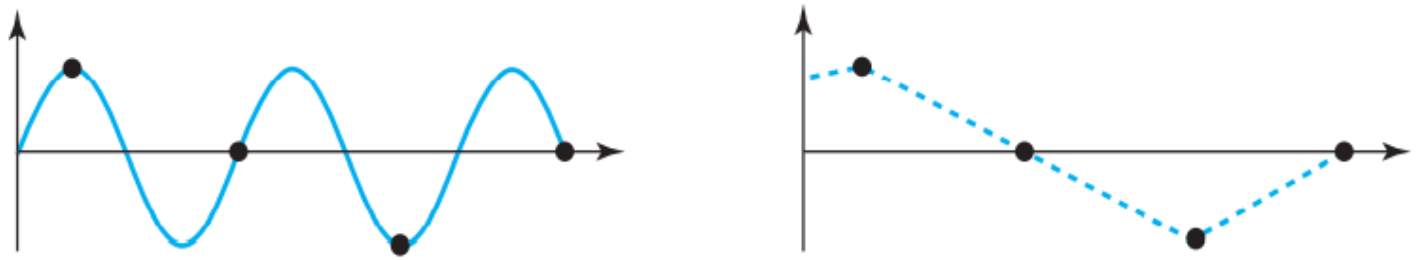
Recovery of a sampled sine wave for different sampling rates



a. Nyquist rate sampling: $f_s = 2f$



b. Oversampling: $f_s = 4f$



c. Undersampling: $f_s = f$

QUANTIZATION

- The result of sampling is a series of pulses with amplitude values between the maximum and minimum amplitudes of the signal.
- The set of amplitudes can be infinite with nonintegral values between the two limits.
- These values cannot be used in the encoding process.

STEPS IN QUANTIZATION

- We assume that the original analog signal has instantaneous amplitudes between V_{min} and V_{max}.
- We divide the range into L zones, each of height Δ (delta).

$$\Delta = \frac{V_{min} - Vmax}{L}$$

- We assign quantized values of 0 to L − 1 to the midpoint of each zone.
- We approximate the value of the sample amplitude to the quantized values.

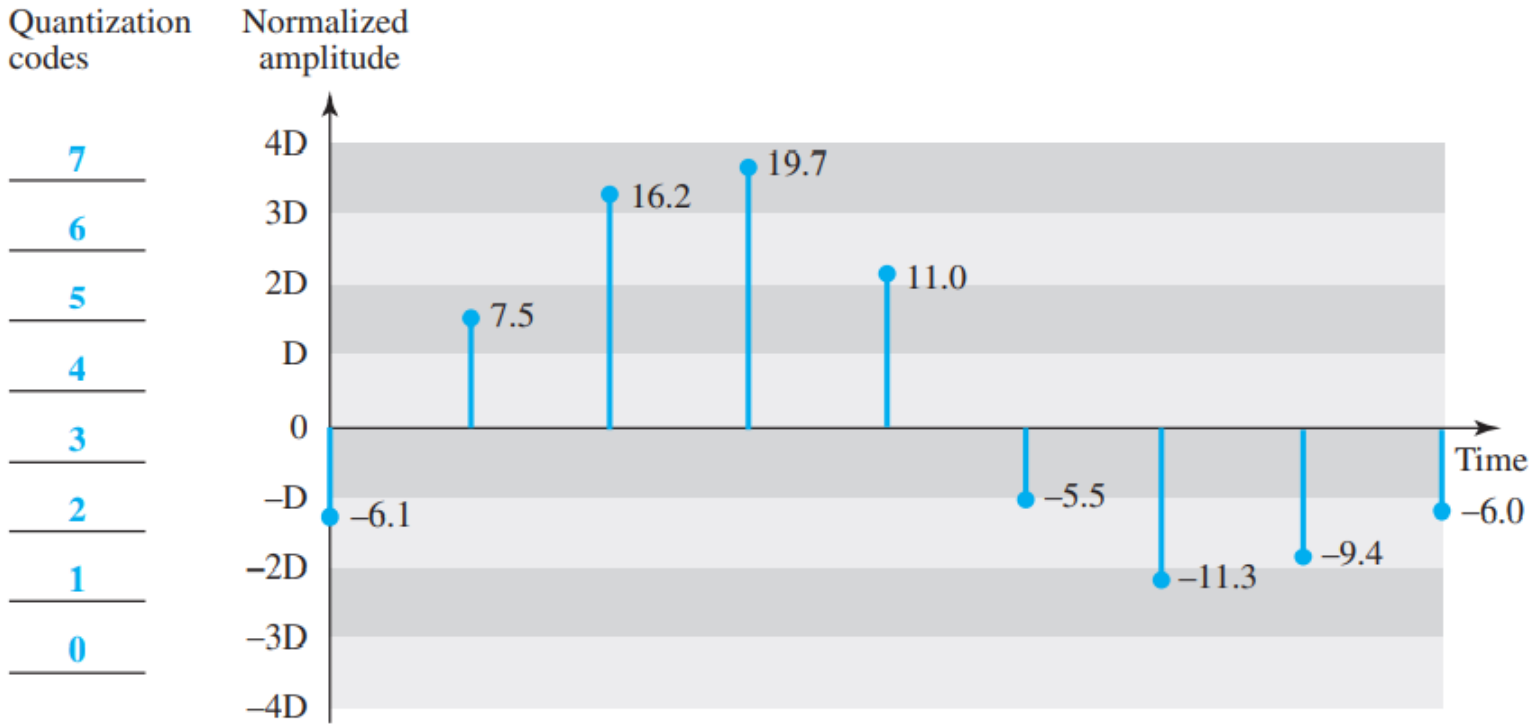
CONTD…

- Quantization Levels
- Quantization Error
  - One important issue is the error created in the quantization process
  - Quantization is an approximation process.

CONTD…

- The quantization error to the SNR_{dB} of the signal depends on the number of quantization levels L, or the bits per sample n_b, as shown in the following formula:

$$SNR_{dB} = 6.02n_b + 1.76 \text{ dB}$$



Normalized PAM values	–1.22	1.50	3.24	3.94	2.20	–1.10	–2.26	–1.88	–1.20
Normalized quantized values	–1.50	1.50	3.50	3.50	2.50	–1.50	–2.50	–1.50	–1.50
Normalized error	–0.38	0	+0.26	–0.44	+0.30	–0.40	–0.24	+0.38	–0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

PROBLEM – ONE MINUTE PAPER

- What is the SNR_{dB} for the example shown?

TYPES OF QUANTIZATION

- Uniform Quantization
- Non-Uniform Quantization

ENCODING

- After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n_b-bit code word
- A quantization code of 2 is encoded as 010; 5 is encoded as 101; and so on.
- The number of bits for each sample is determined from the number of quantization levels.
- If the number of quantization levels is L, the number of bits is $n_b = \log_2 L$. In our example L is 8 and n_b is therefore 3.

Bit rate = sampling rate × number of bits per sample = $f_s \times n_b$

EXIT CLIPS

- We want to digitize the human voice. What is the bit rate, assuming 8 bits per sample?

  (Human voice - frequencies from 0 to 4000 Hz)

f_m

$f_s = 2 f_m$

$f_s = 8000\ Hz$

Bit rate = Sampling rate $\times n_b$

$= f_s \times n_b$

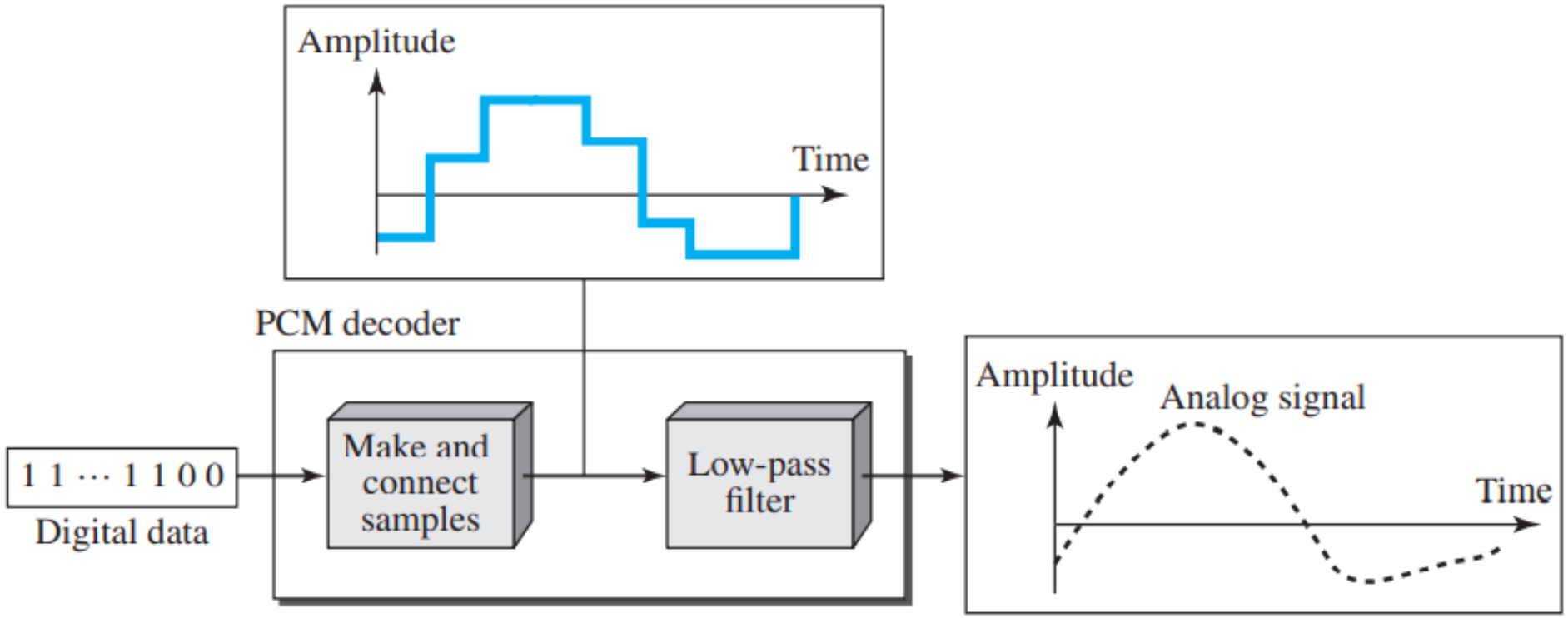
$= 8000 \times 8 = 64000 = 64 \times 10^3$

Bit rate = 64 Kbps

ORIGINAL SIGNAL RECOVERY

- The recovery of the original signal requires the PCM decoder. The decoder first uses circuitry to convert the code words into a pulse that holds the amplitude until the next pulse.
- After the staircase signal is completed, it is passed through a low-pass filter to smooth the staircase signal into an analog signal.
- The filter has the same cutoff frequency as the original signal at the sender.
- If the signal has been sampled at (or greater than) the Nyquist sampling rate and if there are enough quantization levels, the original signal will be recreated.

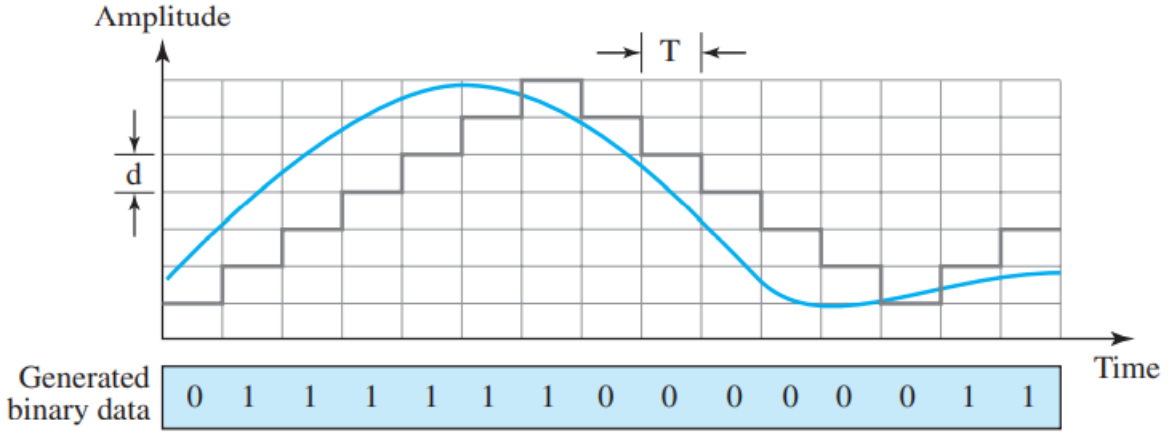
Components of a PCM decoder



DELTA MODULATION (DM)

INTRODUCTION

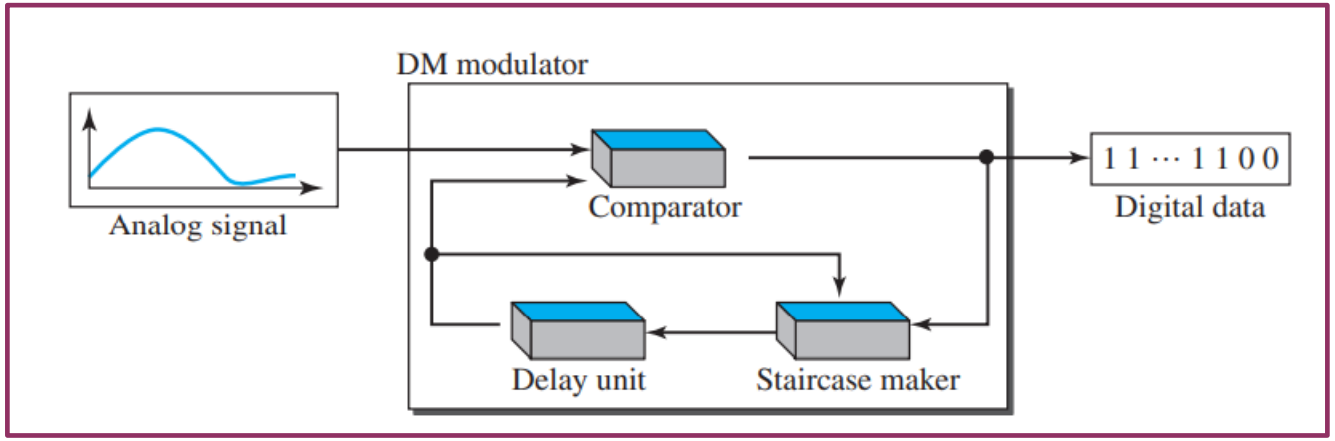
- PCM is a very complex technique. The simplest is delta modulation.
- PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample.

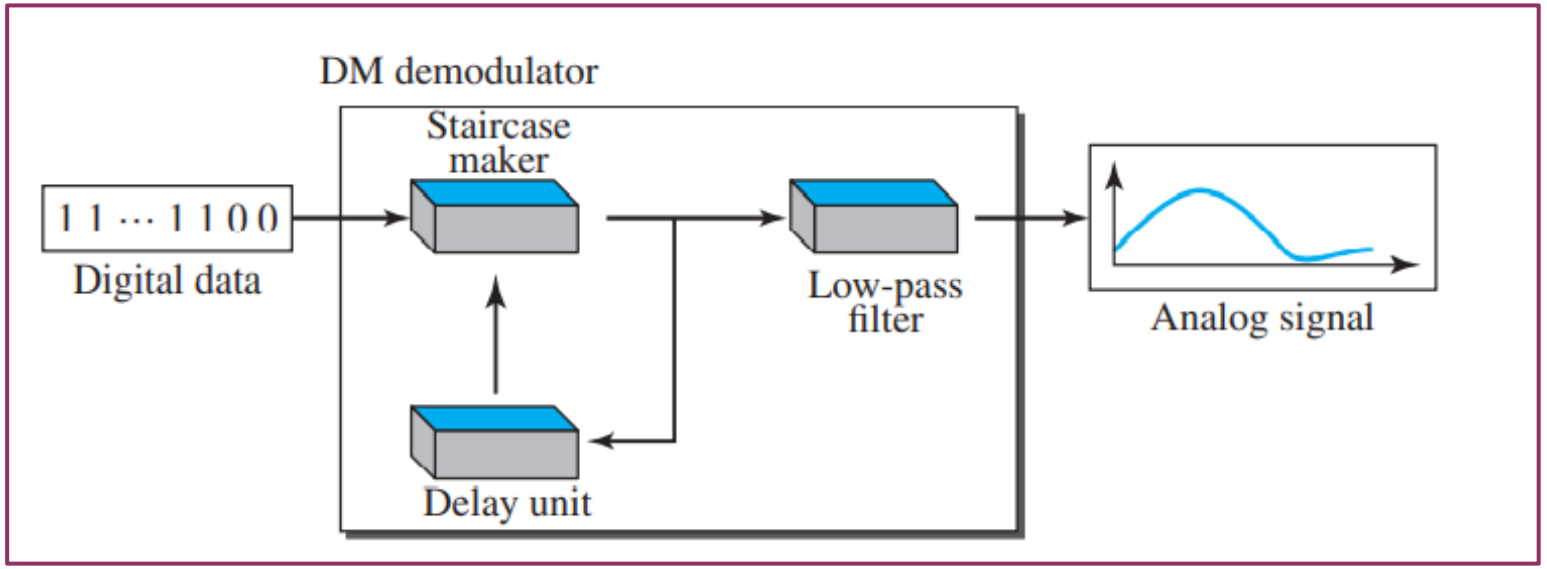


CONTD…

- Modulator
- Demodulator
- Adaptive DM

Delta Modulation & demodulation components





MODULATOR

- The modulator is used at the sender site to create a stream of bits from an analog signal.
- The process records the small positive or negative changes, called delta δ.
- If the delta is positive, the process records a 1; if it is negative, the process records a 0.
- The modulator builds a second signal that resembles a staircase.
- Finding the change is then reduced to comparing the input signal with the gradually made staircase signal.

CONTD…

- The modulator, at each sampling interval, compares the value of the analog signal with the last value of the staircase signal.
- If the amplitude of the analog signal is larger, the next bit in the digital data is 1; otherwise, it is 0.
- The output of the comparator, makes the staircase.
- If the next bit is 1, the staircase maker moves the last point of the staircase signal δ up; if the next bit is 0, it moves it δ down.
- A delay unit is required to hold the staircase function for a period between two comparisons.

DEMODULATOR

- The demodulator takes the digital data and, using the staircase maker and the delay unit, creates the analog signal.
- The created analog signal, however, needs to pass through a low-pass filter for smoothing.

ADAPTIVE DM

- A better performance can be achieved if the value of δ is not fixed.
- In adaptive delta modulation, the value of δ changes according to the amplitude of the analog signal.
- The quantization error of DM, is much less than that for PCM.